

DL Sprint 4.0 | BUET CSE FEST 2026

Luck is all you need

Team Members



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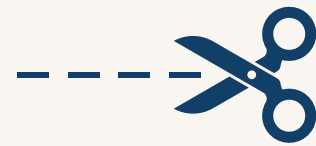
Sk. Ashrafuzzaman Nafees
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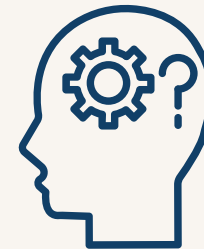
Task 1: Bengali Long-form Speech Recognition

Challenges



Where do I cut?

113 recordings, each ~1 hour long. Only paragraph-level annotations — no timestamps, no boundaries. But whisper needs ≤ 30 s segments.



Whisper hallucinates

Hard cuts mid-word
↓
context loss
↓
garbage output



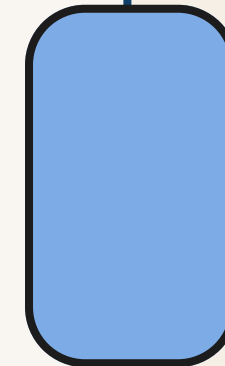
No Low-Resource Tools

No phoneme dictionaries. No pre-built aligners. No annotated boundary datasets.

WhisperAlign: Our Solution

Three challenges. One pipeline.

We built a self-contained, word-boundary-aware chunking pipeline using Whisper's own attention heads — no external tools needed.



Exploratory Data Analysis (EDA)

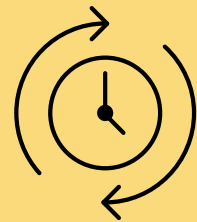
Long-form Bengali audio — no utterance boundaries, no timestamps

Train Dataset



113 recordings

(1 corrupted — train_089.wav)



108.64 hours total

(Avg 57.68 min/.wav)



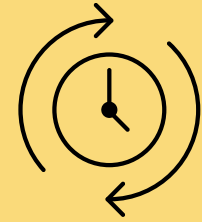
650,065 total words

(Avg 5,753 words/.txt)

Test Dataset



24 recordings



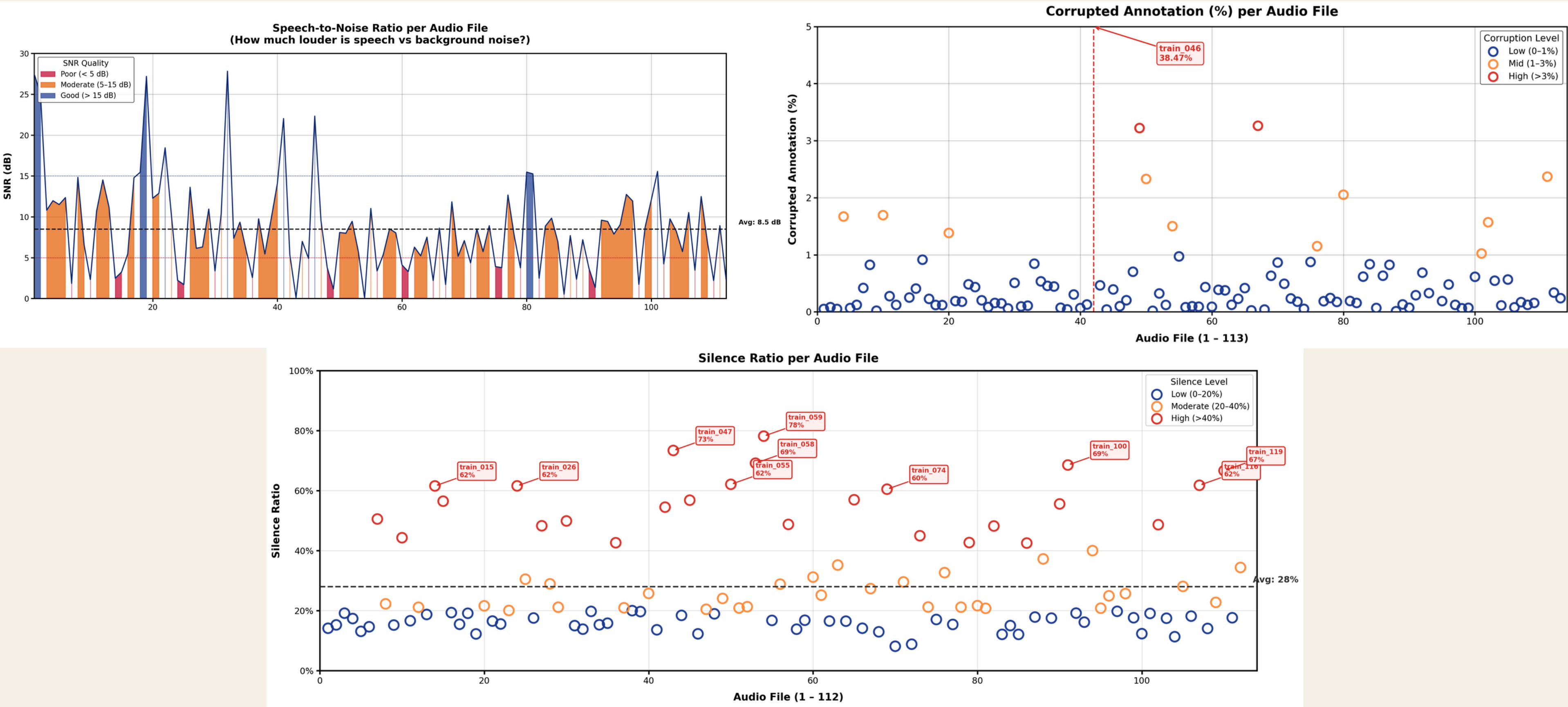
22.20 hours total

(Avg 55.49 min/.wav)



No annotations

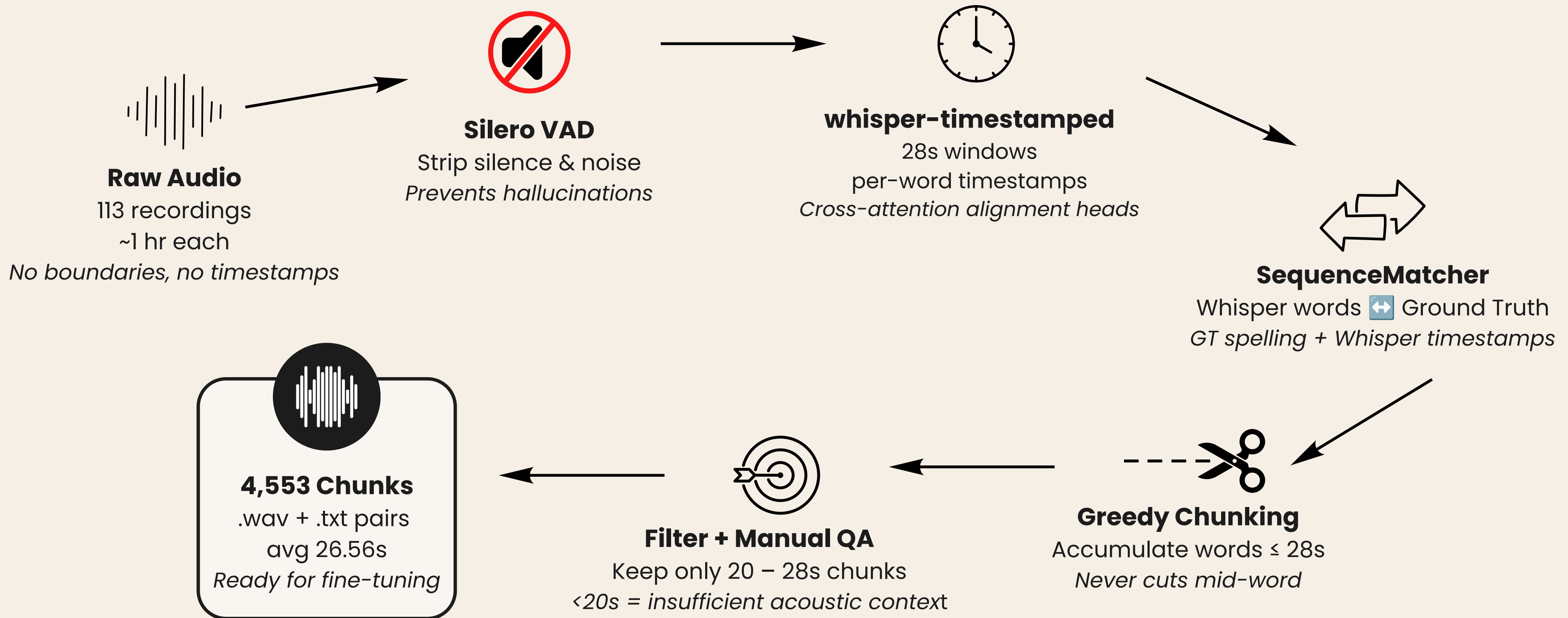
Exploratory Data Analysis (EDA)



Model Selection

Model name	Backbone	Parameter	WER(Public)
hanx	Facebook wav2vec2-xls-r	1 B	0.46932
tugstugi	OpenAI Whisper-medium	763.9M	0.41857
qdv206	ai4bharat/indicwav2 vec_v1_bengali	315M	0.46680
titu_stt_bn_fastcon former + kenLM	NeMo FastConformer	120M	0.54363

Data Preprocessing



Training Configuration

Pre-Training Text Normalization

NFC Unicode Normalization

Canonical decomposition then composition – eliminates invisible encoding inconsistencies

Non-Bengali Characters Removed

Keep only Bengali block (U+0980–U+09FF), ZWJ/ZWNJ, digits, hyphens

Punctuation Stripped

Bengali full stop (।), commas, question marks

Hyperparameters

Epochs
5

Learning Rate
1e-5

Batch Size
16 x 2
= 32 effective

Optimizer
AdamW

Warmup Steps
350

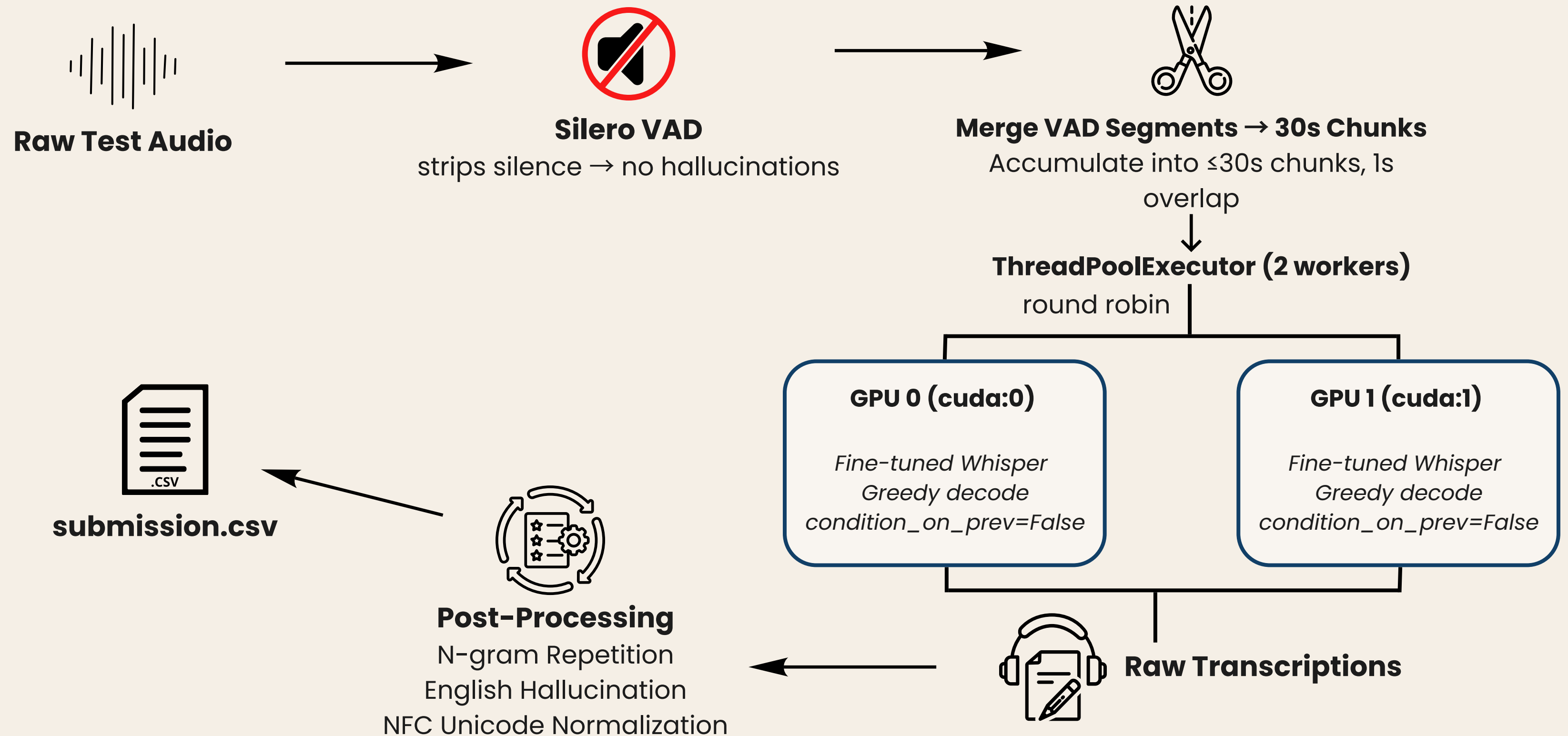
Sampling Rate
16 kHz

Base Model

bengaliAI/tugstugi_bengaliai-asr_whisper-medium

Before WER calculation — NFC normalization applied to both prediction and ground truth

Inference Pipeline



Evaluation Results



WER(Public)
0.25155



WER(Private)
0.27759

System	WER(Public)	WER(Private)
tugstugi — raw, no processing	0.675	0.702
+ VAD + post-processing	0.419	0.440
+ Unicode normalization	0.348	0.375
+ Fine-tuned (our chunking strategy)	0.265	0.296
+ Manual data cleaning (final)	0.251	0.278

Findings & Analysis

Batch Size	WER	RTF	Inference Time (Dual T4 GPU)
1 <i>2 effective</i>	0.251	0.135	3 hrs
8 <i>16 effective</i>	0.269	0.034	46 min

Common Error Patterns

Banglish Words

Model struggles to transcribe banglish words e.g.
welcome → ওয়াকাম, excuse → এক্সকিউর

Echo Sensitivity

Model performance degrades noticeably on recordings with echo. Hallucinations increase when speech boundaries blur.

Task 2: Bengali Long-form Speaker Diarization

Goals

~1 hour recordings, ~22 speakers, real conversational Bangla speech.

Goal: mutually exclusive speaker-timestamped output

Western Acoustic Priors

Existing diarization models are trained on English/European speech and fail to capture Bangla conversational prosody and turn-taking patterns.

Strict Non-Overlap Requirement

Competition rules require mutually exclusive speaker tracks — standard multi-label pipelines produce overlapping segments that must be resolved.

Our Solution - Overview

We domain-adapt Pyannote for Bangla by fine-tuning its segmentation model and introducing a Dual-VAD intersection — achieving ~54% DER reduction

**Pyannote 3.1 Baseline
DER**

42.2%

out-of-the-box

Our Final DER

19.4%

public leaderboard

Reduction

~54%

relative reduction

Key Changes

01.

Bangla-Adapted Segmentation Fine-Tuning

Fine-tune only the 1.4M-param segmentation head on Bangla audio. Embedding model stays frozen.

02.

Dual-VAD Intersection (WhisperX $\cap$ Pyannote)

AND-mask Pyannote's speaker clusters against Silero VAD boundaries so timestamps from both pipelines stay in sync.

Model Selection

Benchmark Dataset	Domain	Pyannote 3.1 DER	Community-1 DER
AHI (IHM)	Meetings	18.8%	17.0%
DIHARD 3 (full)	Diverse	21.4%	20.2%
CALLHOME (p2)	Telephone	28.5%	26.7%
VoxConverse (v0.3)	Web Video	11.2%	11.2%

Why Community-1 ?

01.

Lower DER Across the Board

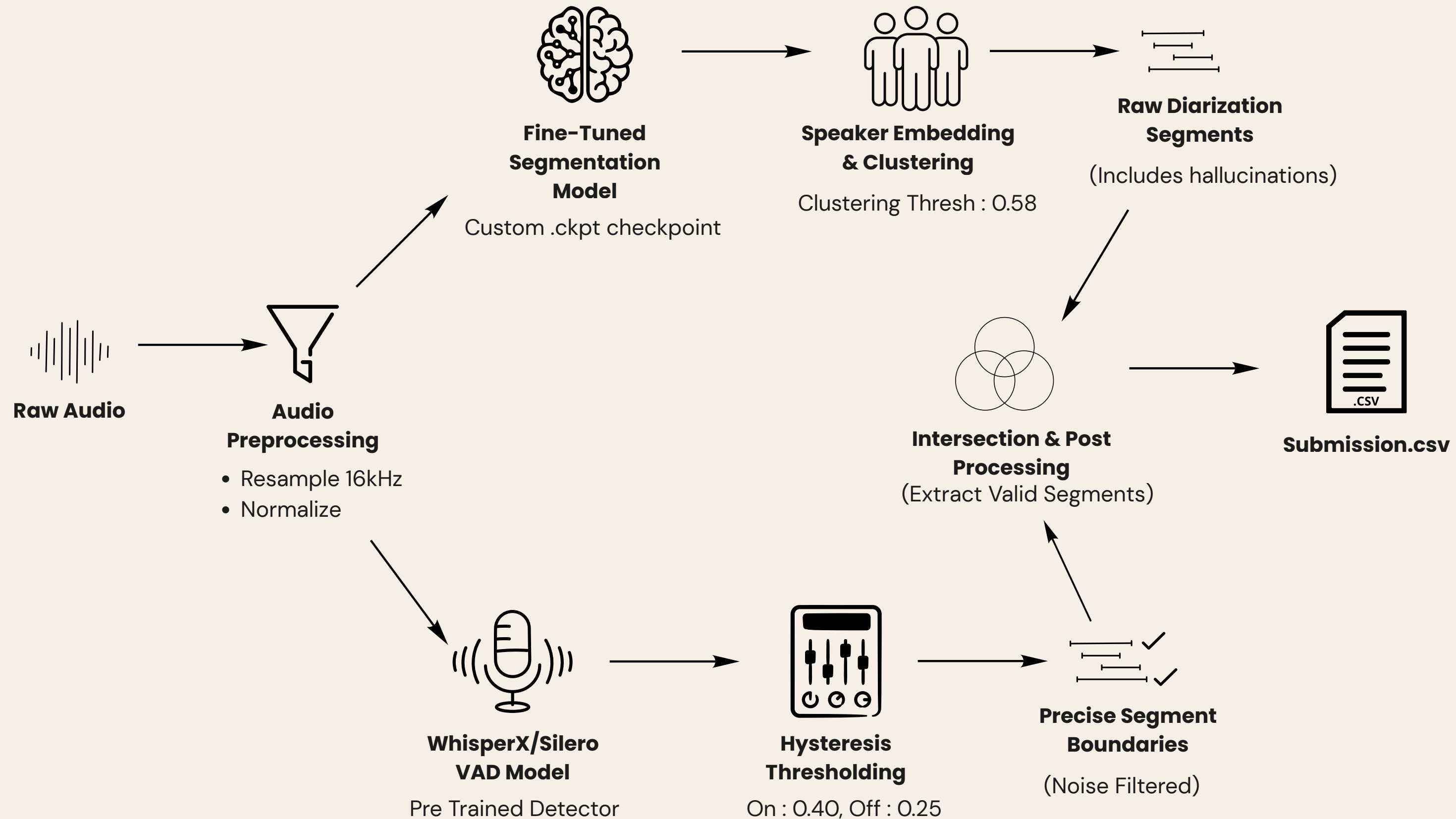
Outperforms Pyannote 3.1 on meetings, diverse audio, and telephony benchmarks – before any fine-tuning.

02.

exclusive_speaker_diarization

Natively resolves overlapping speech by assigning each frame to the dominant speaker – no manual deletion needed

Proposed parallel diarization architecture



Pipeline Configuration

Optimal Hyperparameters

Clustering Threshold
0.58

min_duration_off
0.05 s

VAD onset
0.40

VAD offset
0.25

Pre-Processing

Resample to 16 kHz mono

Peak normalize : $\max * 0.95$

Post-Processing

Drop Segments < 0.15s

Merge : Base gap 0.4s

Adaptive merge : clamped [0.15s – 0.8s]

Fine-Tuning the Segmentation Model

Why Only the Segmentation Model?

- Segmentation = when speakers change → Bangla-specific, must adapt.
- Embedding = who is speaking → cross-lingual, already generalizes.
- 1.4M of 8M params trained.

Base Model
pyannote/segmentation-3.0

Trainable Params
~1.4M (frozen : 6.6M)

Training Data
10 audio files (80/20 split)

Epochs
30 (converges in ~12)

Hardware / Time
Kaggle T4 GPU | ~1 hour

Learning Rate / Batch size
 $1 \cdot 10^{-4}$ | 8 (chunk: 10s)

Impact of Segmentation Fine-tuning

Progression of Evaluation Metrics

System	Public DER	Private DER	Change (Public DER)
Pyannote 3.1 Baseline	42.2%	47.7%	-
+ Preprocessing	36.4%	41.3%	-5.8pp
→ Community-1 Base	31.4%	37.6%	-5.0pp
→ Fine-tuned Segmentation	20.0%	26.1%	-11.4pp ↓

Dual-VAD Intersection

Two pipelines, two VAD models — their speech boundaries don't agree.

01.

WhisperX runs Silero VAD

Silero marks speech/silence boundaries (onset=0.40, offset=0.25)

02.

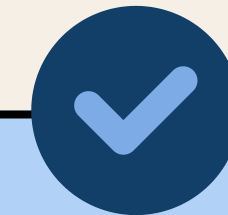
Pyannote clusters independently

Uses its own VAD – boundaries don't match Silero's

03.

AND-mask : the fix

Keep only Pyannote segments that overlap with a Silero speech region



Impact of Dual-VAD Intersection

System	Public DER	Private DER	Change (Public DER)
Pyannote 3.1 Baseline	42.2%	47.7%	-
+ Preprocessing	36.4%	41.3%	-5.8pp
→ Community-1 Base	31.4%	37.6%	-5.0pp
→ Fine-tuned Segmentation	20.0%	26.1%	-11.4pp ↓
→ + Dual-VAD Intersection (final) ★	19.4%	26.0%	-0.6pp ↓

-22.8 pp

Total Public DER
drop

19.4%

Best public DER

~1 HOUR

T4 GPU training

Findings and Analysis

- **Clustering threshold is non-linear** : 0.58 outperformed 0.605 and its neighbours.
- **Content type affects output**: test_012 had only 7 segments — likely a single-speaker epic recitation.
- **Native exclusivity beats manual overlap removal**: Preserves acoustic data; manual deletion destroys it.

Summary

01.

Bangla-Adapted Segmentation Fine-Tuning

Fine-tune only the 1.4M-param segmentation head on Bangla audio. Embedding model stays frozen.

02.

Dual-VAD Intersection (WhisperX $\cap$ Pyannote)

AND-mask Pyannote clusters against Silero VAD. Fixes timestamp mismatch.

19.4%

final Public DER

(Private : 26.0%)

~54%

DER Reduction

From 3.1 Baseline (42.2 %
DER)

1h 25m

Inference Time

Full test set on P100

~1 hr

Training Time

T4 GPU: 20-30 epochs

Thank you

